

**Explainable AI (XAI) Handout:
Demystifying 1D-CNN Classification & Gradient
Saliency Mapping**

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Contents

1	Introduction: The “Black Box” Problem in Spectroscopy	2
2	Conceptual Architecture of a 1D Convolutional Layer	2
2.1	1. The Convolution Operation	2
2.2	2. Non-Linear Feature Extrapolations (ReLU)	2
3	Explainable AI: Deriving the Saliency Map	3
3.1	1. Mathematical Foundation	3
3.2	2. Code Implementation & Normalization	3
4	Interpreting Laboratory Diagnostic Figures	3
4.1	1. Grid Alignment and Resolution Fix	3
4.2	2. Deciphering the Overlaid Heatmap Points	4

1 Introduction: The “Black Box” Problem in Spectroscopy

When using advanced characterization tools like Fourier-Transform Infrared (FTIR), X-ray Diffraction (XRD), or Raman spectroscopy, researchers frequently seek to classify material states (e.g., separating “Pristine” controls from environmentally “Degraded” or contaminated samples). While traditional methods rely on manual baseline corrections, peak integration, or peak-ratio rules, deep learning classifiers can automatically capture minute shifts in complex raw spectral data.

However, standard deep neural networks often behave like a “black box,” making highly accurate predictions without revealing their analytical reasoning. For an experimental scientist, a prediction is legally and academically insufficient without mathematical verification. We must know why a model made its decision to ensure it tracks physical chemistry rather than baseline instrument drift or random background noise.

This reference manual explains the underlying mathematical and physical principles behind the automated workflow in **Module 5-1**, demonstrating how Gradient Tape Saliency Mapping turns a black-box model into a fully verifiable chemical diagnostic tool.

2 Conceptual Architecture of a 1D Convolutional Layer

Unlike standard computer vision architectures that process 2D spatial pixel matrices (width $\times$ height), spectroscopic classifiers handle 1D arrays where the horizontal axis corresponds to a continuous sequential domain (time steps, coordinates, or wavenumbers).

2.1 1. The Convolution Operation

Instead of evaluating every data point independently, a convolutional layer slides a small localized window—known as a kernel—across the continuous spectral sequence. The code initializes this operation using:

```
1 layers.Conv1D(32, 15, activation='relu', input_shape=(500, 1))
```

This translates to 32 independent analytical filters, each spanning a width of 15 sequential data channels. Mathematically, at any point t along the input timeline, the layer calculates a localized dot product:

$$y(t) = \sum_{i=0}^{K-1} x(t+i) \cdot w(i) + b \quad (1)$$

where x represents the raw signal amplitude, w is the localized filter weight array assigned by the network, K is the kernel width (15 steps), and b is a scalar correction bias. This mimics the way a chemist scans a spectrum for specific structural peak envelopes or side-band anomalies.

2.2 2. Non-Linear Feature Extrapolations (ReLU)

To capture complex chemical variances, the linear output of the convolution is transformed by the Rectified Linear Unit (**ReLU**) activation function:

$$f(y) = \max(0, y) \quad (2)$$

ReLU replaces negative mathematical outcomes with zero while preserving positive features. This filters out background noise and passes sharp spectral peaks or shoulder features to the next layer in the sequence.

3 Explainable AI: Deriving the Saliency Map

Once the network achieves high validation accuracy, we use Explainable AI (XAI) to verify its internal logic. The script uses a custom Gradient Tape tracking loop to calculate how a tiny change in a specific wavelength or data point affects the final model prediction.

3.1 1. Mathematical Foundation

Let $S_c(X)$ represent the continuous output score computed by the trained neural network for a target category c (e.g., classifying a spectrum as “Degraded”) given an input signal profile vector $X = [x_1, x_2, \dots, x_{500}]$.

To measure how much an individual data coordinate x_i influences the final classification, the pipeline calculates the partial derivative of the prediction score relative to that specific input step:

$$G_i = \frac{\partial S_c(X)}{\partial x_i} \quad (3)$$

This partial derivative measures the model’s sensitivity at that precise point. A steep gradient means that a tiny shift in that specific peak’s intensity strongly changes the final classification score, proving that the network is actively focusing on that region.

3.2 2. Code Implementation & Normalization

The tracking loop monitors this calculation through the following Python sequence:

```
1 with tf.GradientTape() as tape:
2     tape.watch(img_tensor)
3     prediction = model(img_tensor)
4
5 # Extract the true partial derivatives via backpropagation
6 gradients = tape.gradient(prediction, img_tensor)
7 saliency = np.abs(gradients.numpy()).squeeze()
```

Because a gradient can take both positive and negative values depending on the direction of the signal change, the pipeline takes the absolute value (`np.abs`) to measure absolute structural importance. It then rescales the array from 0.0 to 1.0 to create a standardized importance metric:

$$\text{Saliency}_{\text{Norm}}(i) = \frac{G_i - \min(G)}{\max(G) - \min(G)} \quad (4)$$

4 Interpreting Laboratory Diagnostic Figures

The script outputs a multi-panel visualization designed to provide peer-reviewed validation for your research papers.

4.1 1. Grid Alignment and Resolution Fix

The visualization engine enforces a strict structural update to prevent layout mismatches:

```
1 fig, axes = plt.subplots(2, 1, figsize=(12, 7), sharex=True)
```

Setting `sharex=True` links the horizontal axes of both plots perfectly. This ensures that every raw spectral peak on the top plot aligns precisely with its corresponding attention weight below, making it easy to identify key features by eye.

4.2 2. Deciphering the Overlaid Heatmap Points

- **The Background Curve:** Displays the raw input signal from an experimental sample flagged as “Degraded”.
- **The Scatter Overlays:** Change color dynamically based on the calculated saliency array. High-importance regions flare into bright yellow and white points, while uninformative background regions remain cool dark red.

If your material contains a high-frequency contaminant shoulder, the scatter markers will light up exclusively over that artifact region. This provides a definitive visual proof for your reports that the model has successfully identified the correct physical features rather than relying on baseline anomalies.